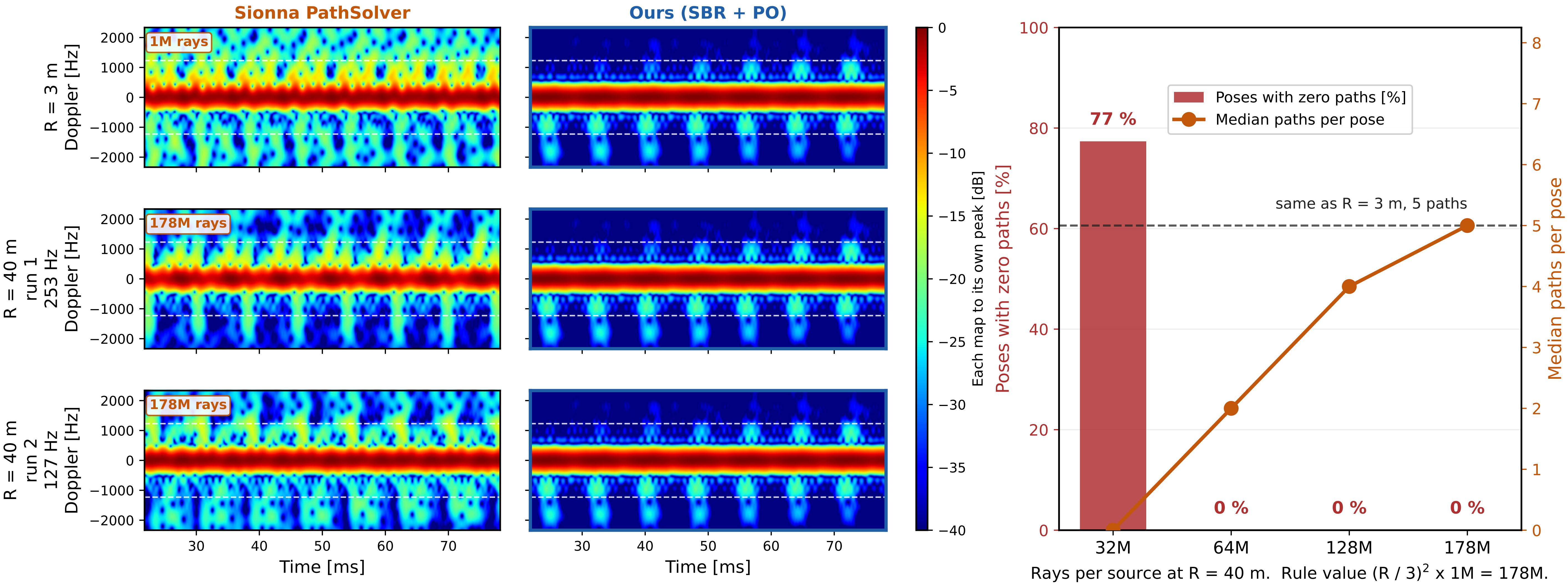
DJI Matrice 4E, belly view, 3.5 GHz. Left, one row per range, the same 60 ms window. Right, what it costs Sionna to reach 40 m.



Left. Only the Sionna column is solved again at each range, with the ray budget raised as the target shrinks in solid angle. Our column is computed once and reused, because a plane wave anchored on the target carries no range at all. That repetition is the claim, not an accident of plotting. No pose is ray starved at any range. Maps are normalized to their own peak, so this compares structure only, never level.

The two 40 m rows are the same computation with a different random seed. The level agrees within 2.6 dB and the strongest modulation line does not, at 253 Hz and 127 Hz against a prediction of 127 Hz. Our column has no seed in it at all.

Right. At 40 m a starved budget leaves most poses with no path at all, which looks like a collapse. Raising the ray count alone removes it, and at the rule value the path count matches the 3 m case. A separate run at the same 32M budget gave 78 % empty poses, close to the 32M bar here. Ledger outputs/report07_ray_budget_test.json, 128 poses per rung, 5 kHz.